**MGU-UGP BSc (HONS) MICROBIOLOGY**

**IV SEMESTER**

# DSC C - MG4DSCMBG202 – APPLIED MICROBIOLOGY

**(2024 ONWARDS)**

**Time: 1.5 Hours** **Max. marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

## Part A

**Short Answer Type Questions**

**Answer any 10 questions**

**Each question carries 2 marks**

1. Name any two useful microbes in the food industry. [K][1]
2. What is Tempeh? [K][1]
3. Give a note on MBRT test? [U][1]
4. Define nif gene. [K][2]
5. Explain about fermentation. [U][1]
6. Write a short note on commensalism. [K][2]
7. Explain about biopesticides. [U][2]
8. What is meant by upwelling in an aquatic ecosystem? [K][3]
9. What is the role of microbial indicator in water pollution? [K][3]
10. Describe about settle plate method. [U][4]
11. List out air borne bacterial infections. [K][4]
12. Mention the role of benthic microorganisms in marine ecosystem. [K][4]

**(10x2=20)**

## Part B

**Short Essay Questions**

**Answer any 4 questions**

**Each question carries 5 marks**

13. Explain production of edible mushroom. [U][1]
14. Write a note on Probiotics and its significance. [K][1]
15. What is biopesticides? What are the benefits of biopesticides. [U][2]
16. What is an estuary? Give an account on estuarine microflora. [K][3]
17. Discuss three positive interactions of soil microorganisms. [U][2]
18. Describe the different impingement methods of air sampling. [U][4]

**(4x5=20)**

## Part C

**Long Essay Question**

**Answer any 1 question**

**Each question carries 10 marks**

19. Explain MPN method for testing the presence of coliforms in water. [U][3]
20. Distinguish between symbiotic and non-symbiotic nitrogen fixation? [U][2]

**(10x1=10)**